ORF 309 – Homework 4, Question 3 Solution

Problem Statement

Every day, you eat one box of CrunchYum™ breakfast cereal. The manufacturer puts little action figures in the boxes that you like to collect. There are 4 distinct types of action figures. Each box contains an action figure with probability $\frac{1}{2}$; among those boxes that contain an action figure, each type is equally likely. The content of every cereal box is independent.

Suppose you have already collected k distinct types of action figures ($0 \leq k < 4$). What is the expected number of boxes you need to eat until you get an action figure of a type that you do not have yet?

(b) From the time that you first started eating CrunchYum™, what is the expected number of boxes you need to eat until you have collected at least one action figure of every type?

Solution to Part (a)

Let $A =$ the number of boxes needed to get an action figure of a type that you do not have yet, given that you have already collected k distinct types.

Step 1: Compute $P(A = \textbf{success} \mid k)$

A single box gives you a new type if two things happen:

1. The box contains an action figure (probability $\frac{1}{2}$), **and**
2. That figure is one of the $(4 - k)$ types you are missing (probability $\frac{4-k}{4}$, since each type is equally likely).

By independence of these two events:

$$P(A \mid k) \;=\; \frac{1}{2} \cdot \frac{4-k}{4} \;=\; \frac{4-k}{8}.$$

Step 2: Recognize the geometric distribution

Each box is an independent Bernoulli trial with success probability $P(A \mid k)$. Therefore:

$$A \;\sim\; \text{Geom}\left(\frac{4-k}{8}\right).$$

The expectation of a Geom(p) random variable (number of trials until the first success) is $1/p$.

Step 3: Result

$$\boxed{E[A] \;=\; \frac{1}{P(A \mid k)} \;=\; \frac{8}{4-k}}$$

Sanity Checks

k (types collected)	Success prob. p	Expected boxes $1/p$
0	$4/8 = 1/2$	**2**
1	$3/8$	$\mathbf{8/3 \approx 2.67}$
2	$2/8 = 1/4$	**4**
3	$1/8$	**8**

This makes intuitive sense: the more types you have already collected, the harder it is to find a missing one. When $k = 0$, every action figure is new, so you just need to wait for any figure to appear (expected 2 boxes, since each box has a $\frac{1}{2}$ chance of containing one).

Solution to Part (b)

Let A_k denote the number of boxes needed to go from k collected types to $k+1$ collected types. From part (a), $A_k \sim \text{Geom}\big(\frac{4-k}{8}\big)$ with $E[A_k] = \frac{8}{4-k}$.

The total number of boxes to collect all 4 types is:

$$T \;=\; A_0 + A_1 + A_2 + A_3.$$

By **linearity of expectation**, for any random variables $X_1, \ldots, X_n$ (independent or not):

$$E[X_1 + X_2 + \cdots + X_n] \;=\; E[X_1] + E[X_2] + \cdots + E[X_n].$$

Applying this to T:

$$E[T] \;=\; \sum_{k=0}^{3} E[A_k] \;=\; \sum_{k=0}^{3} \frac{8}{4-k} \;=\; \frac{8}{4} + \frac{8}{3} + \frac{8}{2} + \frac{8}{1} \;=\; 2 + \frac{8}{3} + 4 + 8.$$

$$\boxed{E[T] \;=\; \frac{6+8+12+24}{3} \;=\; \frac{50}{3} \;\approx\; 16.67 \text{ boxes}}$$